Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Physics Examination May 2018

Introduction:

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Physics. It contains comments on candidates' responses to the 2018 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes:

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations:

This year candidates performed well on questions related to the law of electromagnetic induction, capacitor and capacitance, AC and DC generators, root mean square value, electromagnetic waves, mechanical properties of solids, law of radioactive decay and special theory of relativity. Whereas low-scoring candidates did not perform well in questions based on current electricity, electric field and electric field intensity, inductance and induction, power dissipation, force on a moving charged particle, semi-conductor devices, atomic spectra, Bohr's model of hydrogen atom, candidates also had problem in articulating their answers on differences, reasoning and discussion type questions.

Detailed Comments:

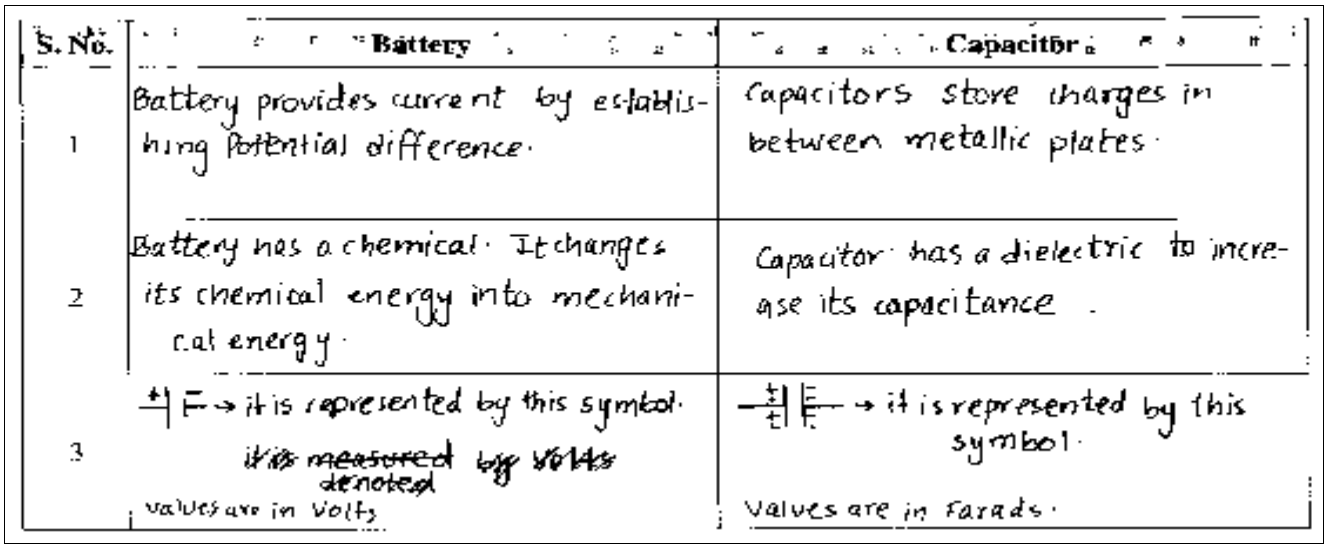
Constructed Response Questions (CRQs)

Question 1:

Compare a battery to a capacitor in THREE points.

Better responses presented correct comparison between a battery and a capacitor in three points. High scoring candidates wrote the comparison in terms of storing chemical and electrical energies, connecting terminals; positive and negative poles, providing direct and alternating currents, production of charges etc.

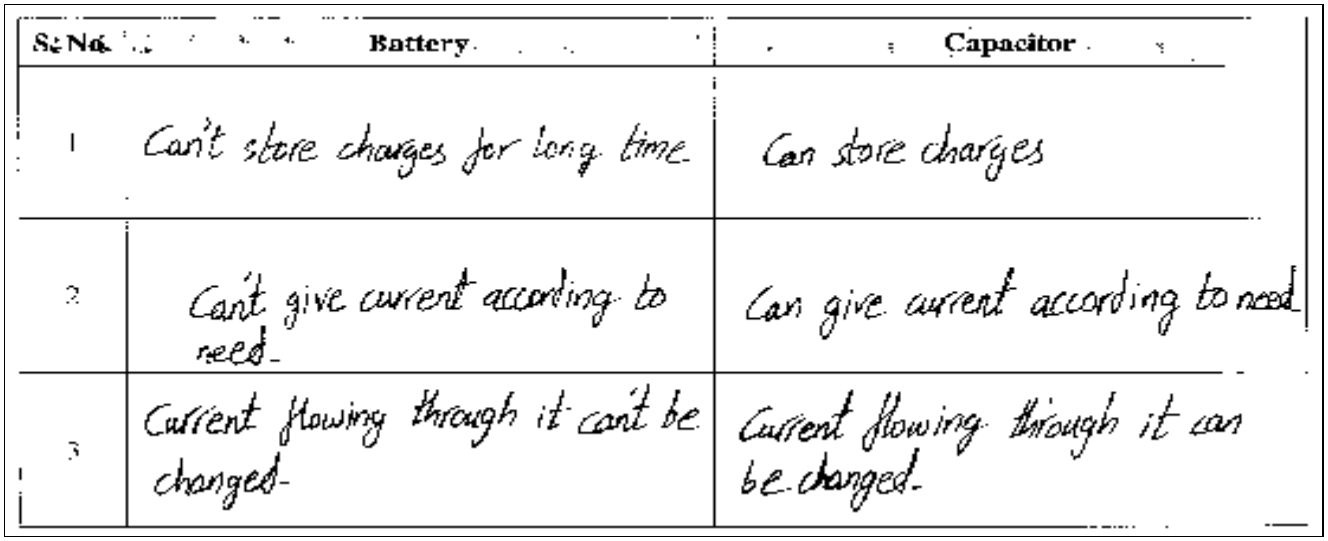
Example:



S. No.	Battery	Capacitor
1	Battery provides current by establishing potential difference.	Capacitors store charges in between metallic plates.
2	Battery has a chemical. It changes its chemical energy into mechanical energy.	Capacitor has a dielectric to increase its capacitance.
3	→ it is represented by this symbol. it is ~~measured~~ denoted by volts. Values are in volts.	→ it is represented by this symbol. Values are in farads.

Weaker responses presented incorrect comparison between a battery and a capacitor. Few of the low scoring candidates wrote the comparisons in negation. Therefore, they were unable to get full marks. They wrote only one or two correct points with correct meaning.

Example:



S. No.	Battery	Capacitor
1	Can't store charges for long time.	Can store charges
2	Can't give current according to need.	Can give current according to need.
3	Current flowing through it can't be changed.	Current flowing through it can be changed.

Question 2:

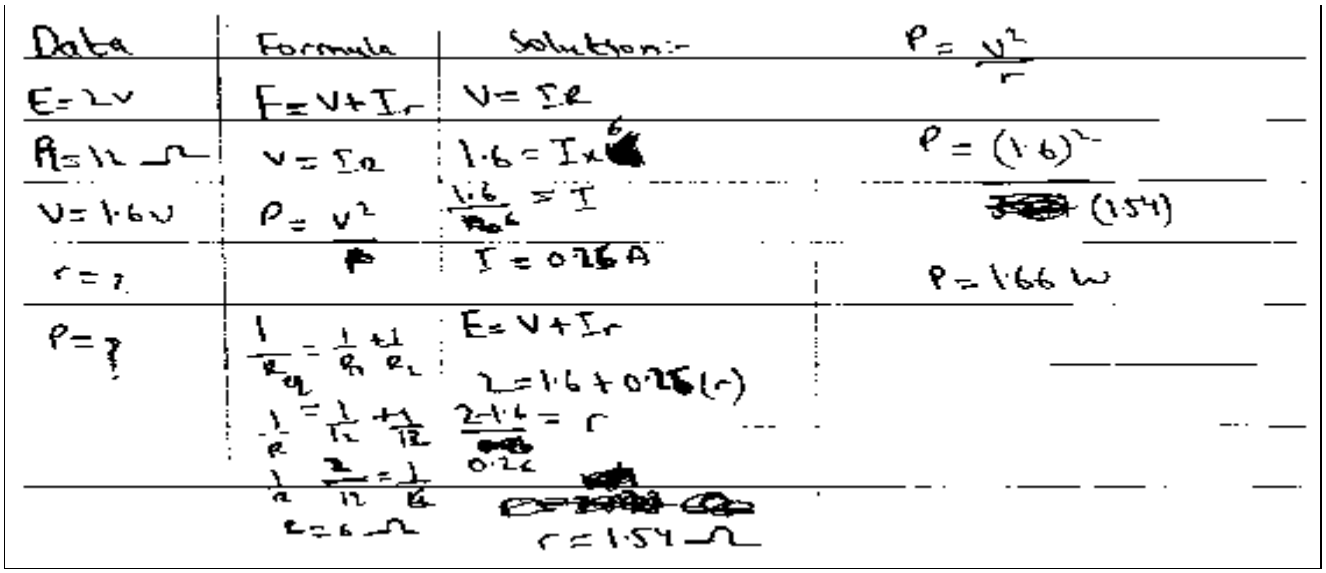
Two identical cells of electromotive force (e.m.f.) 2.0 V are joined together in parallel combination. They provide power to an external circuit consisting of two resistors of 12 Ω each which are joined in parallel combination with a circuit and a voltmeter which reads a voltage of 1.6 V.

Calculate the internal resistance of each cell and the power dissipated by the internal resistance.

Most candidates found this question challenging and hence, the performance was below expectations.

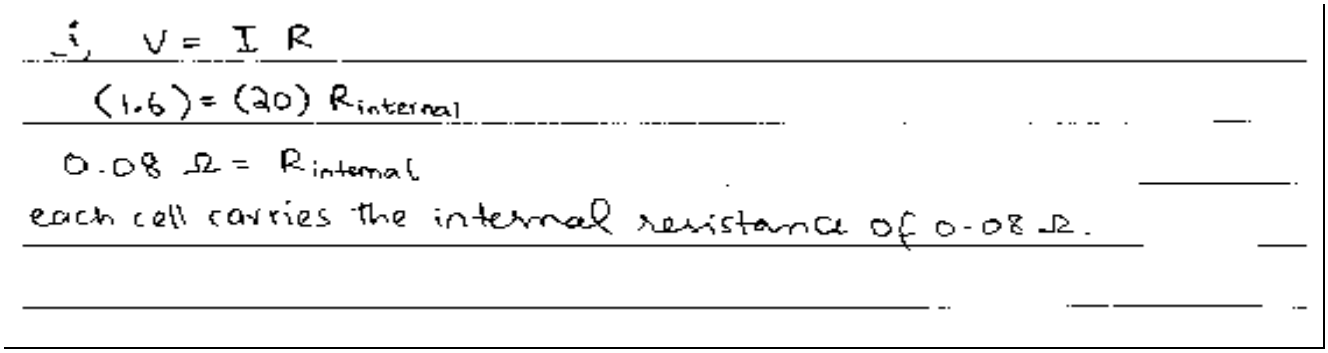
Better responses calculated the internal resistance and power dissipation by putting the correct formula of internal resistance, r = [(E - V) / V] R and calculated the correct answer, i.e, r = 1.5 Ω. High scoring candidates put the correct formula of power dissipation, $P = V \times I = I^2 \times R = \frac{V^2}{R}$ and calculated the correct answer, i.e, P = 1.706 watt.

Example:



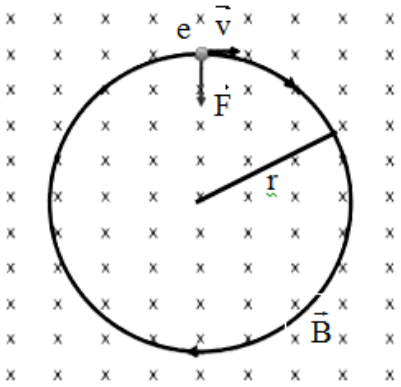
Weaker responses failed to calculate the internal resistance of each cell and the power dissipation by the internal resistance respectively. Most of the low scoring candidates wrote the formula of Ohm's law, V = I R, formula for the effective resistance in series combination, $R_E = R_1 + R_2$ but were unable to apply the formula of power dissipation and internal resistance in their calculations.

Example:



Question 3:

An electron of mass (m) and charge (e) is accelerated with a velocity ($\vec{v}$) by a potential difference (V) which is projected perpendicularly to a uniform magnetic field ($\vec{B}$).



The electron moves along a circular trajectory of radius (r).

a. Mention the expression for the

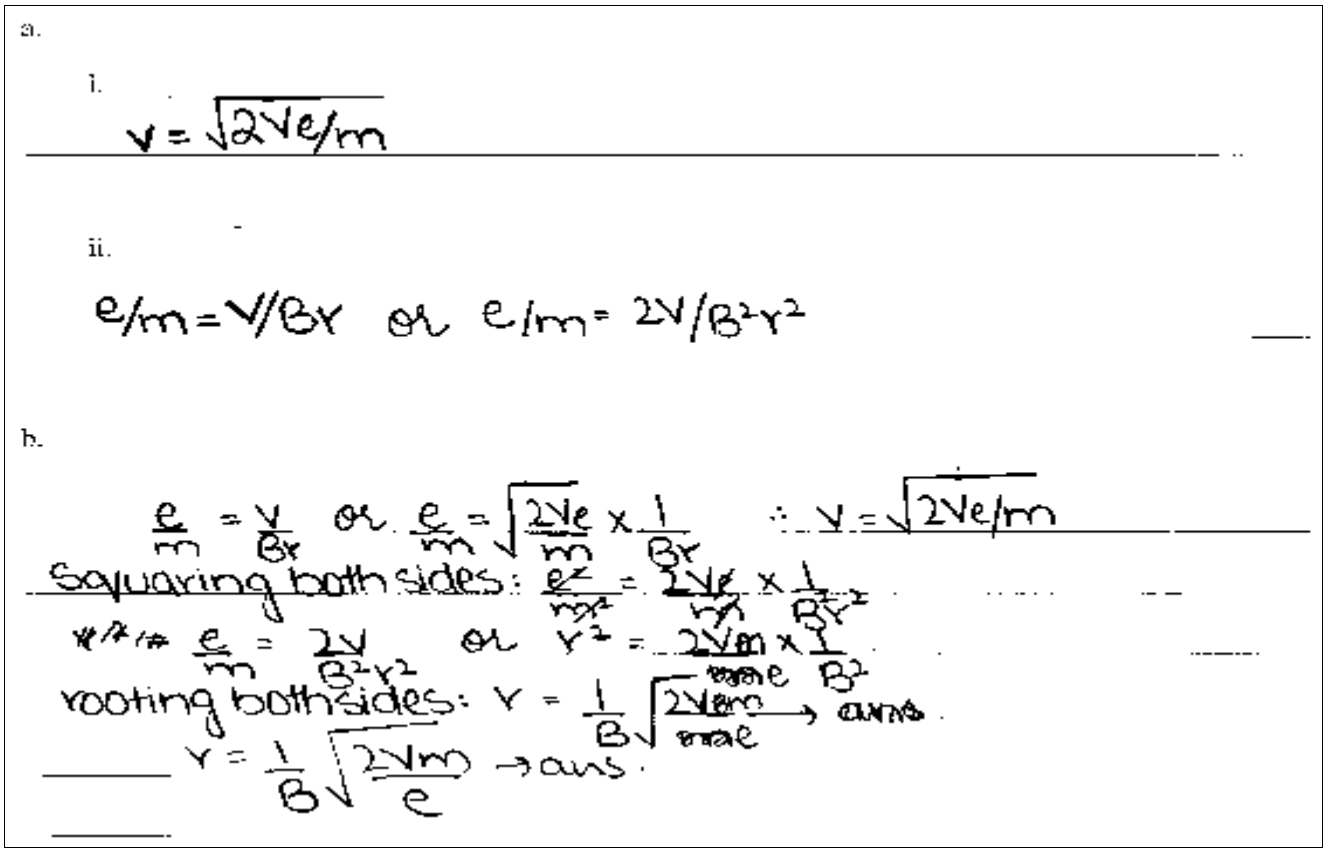
i. speed of the electron.

ii. ratio $\frac{e}{m}$

b. Show that the radius of the circular trajectory of the electron is $r = \frac{1}{B}\sqrt{\frac{2Vm}{e}}$

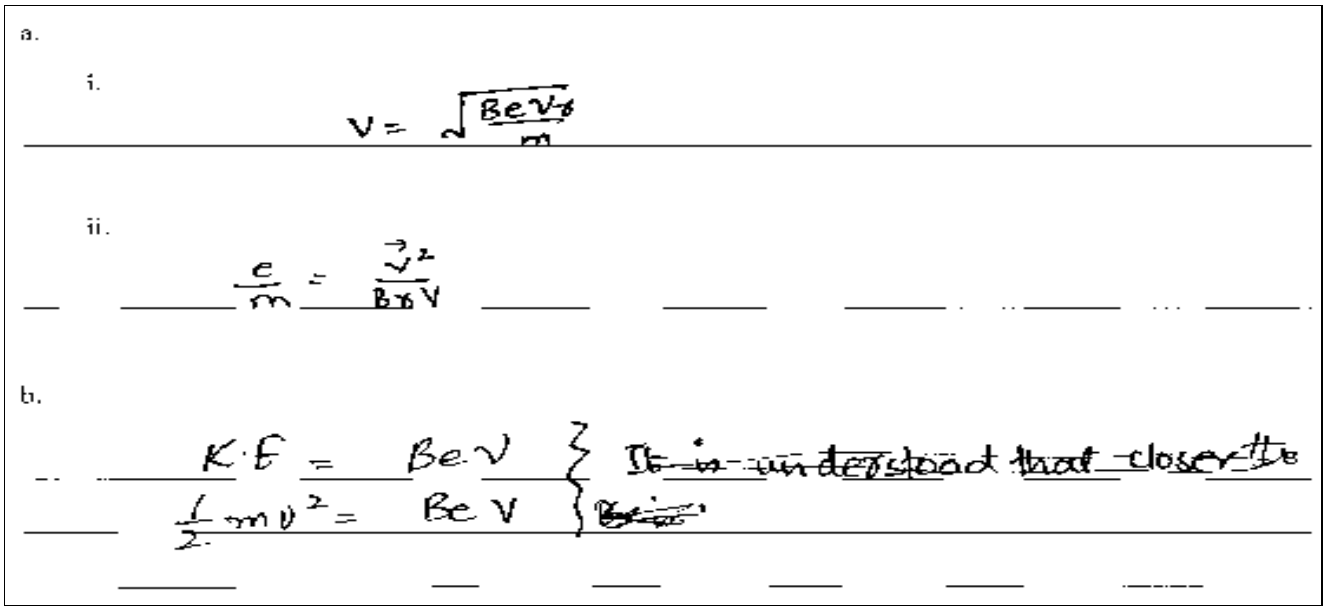
Better responses correctly mentioned the expression for the speed of the electron and ratio of $\frac{e}{m}$ in part (a), i.e, $v = \sqrt{\frac{2Ve}{m}}$ and $\frac{e}{m} = \frac{v}{Br}$ respectively. High scoring candidates also wrote the correct mathematical steps for the radius of the circular trajectory of the electron, i.e, $r^2 = \frac{m^2}{B^2e^2}\left(\sqrt{\frac{2Ve}{m}}\right)^2$, $r^2 = \frac{m}{B^2e}\left(\frac{2V}{1}\right)$, and $r^2 = \frac{2Vm}{B^2e}$.

Example:



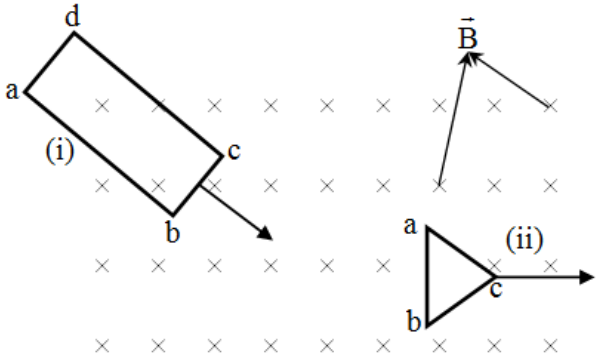
Weaker responses indicated that candidates mentioned the expressions for the speed of the electron and ratio of $\frac{e}{m}$ but were unable to write the correct mathematical steps for the radius of the circular trajectory of the electron. Few of the low scoring candidates wrote the formula of kinetic energy and compared it with the electric field and potential difference of the electron.

Example:



Question 4:

The given figure shows a region of magnetic field ($\vec{B}$).



a. If a rectangular loop (abcd) is entering into the magnetic field, then the magnetic flux (Φ_B) through the rectangular loop (abcd) increases or decreases? Give reason to support your answer.

b. A triangular loop (abc) is leaving the magnetic field. Is the induced current along (bacb) or (bcab)? Give reason to support your answer.

c. If both the loops are completely static inside the region of the magnetic field, then there will be no induced current. True or False?

Better responses wrote the correct reasons in all three parts. High scoring candidates wrote that when more magnetic field lines will pass through the loop, the magnetic flux (Φ) through the rectangular loop (abcd) will increase.

In the second part of the question, candidates wrote that the induced current flows along (bacb) because it opposes the change in flux and has outward motion.

In the last part of the question, candidates wrote that if both the loops are completely static inside the region of the magnetic field, then there will be no induced current at all.

Example:

a.

If the rectangular loop is coming inside the magnetic flux passing through it will increase because of exposure of greater surface area to the magnetic field will increase the magnetic flux passing through it.

b.

The induced current is along the direction (bacb). This is in accordance with lenz law according to which ~~induced current~~ the direction of induced current will always oppose the cause which produces it.

c.

True.

Weaker responses showed the weakness in candidates' understanding of both the rectangular coil and magnetic field and hence, such candidates answered that magnetic flux through the rectangular loop (abcd) 'decreases' when a rectangular loop (abcd) is entering into the magnetic field.

In the last part, candidates wrongly identified that the induced current flows along (abc) or (bcab) and mentioned it as a 'false statement'.

Example:

a.

Magnetic flux through the rectangular loop is decreases when the triangular loop(abc) is connected with rectangular loop (abcd) less magnetic field is passing through them it means less magnetic flux is induced.

b.

when triangular loop leaving the magnetic field it induced current when it connect with abcd it disturb the magnetic flux or the magnetic field.

c.

False.

Question 5:

Define the following terms:

a. Peak value (voltage).
b. Root mean square (r.m.s.) value (voltage).

Most candidates attempted this question well.

Better responses correctly defined peak value (voltage) and root mean square (r.m.s.) value (voltage) by mentioning that the peak voltage refers to the maximum value of the alternating voltage and the root - mean - square value is the square root of $\frac{V^2_o}{2}$.

Example:

a.

The value of the peak when the voltage is maximum. The peak can be positive or negative. At peak value voltage is alway maximum

b.

It is the mean square value of the voltage by taking root.

$V_{rms} = \frac{1}{\sqrt{2}} v_0$ or $V_{rms} = 0.707\, v_0$

Weaker responses reflected that candidates were unable to define the peak value and root mean square value. Few of the low scoring candidates wrote about instantaneous value of alternating current, square root of the value of voltage etc. which are not relevant to the real demand of the question.

Example:

a.

The value which determines the instantaneous value of voltage.

b.

The value which square root of voltage is called root mean square value.

Question 6:

The given diagram shows three types of waves.

1 

2 

3 

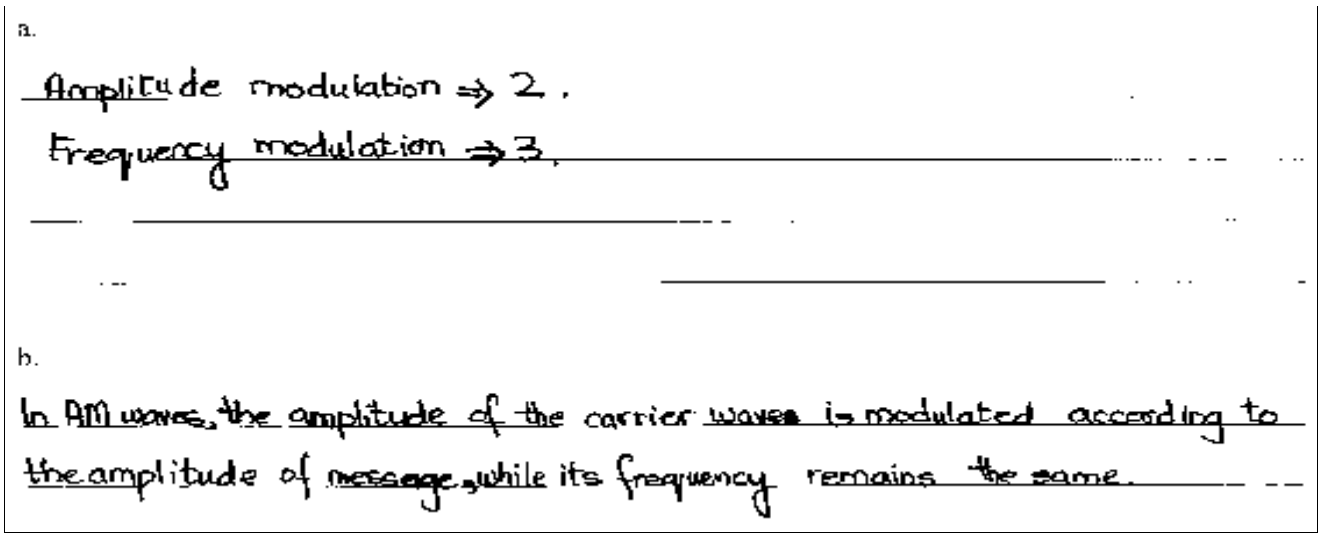
a. Identify amplitude modulation (AM) and frequency modulation (FM) waves.

b. Define any one of the above identified waves.

This question was generally well-attempted by the candidates.

Better responses correctly identified and defined the different types of waves given in the question. High scoring candidates identified wave # 2 as AM wave and wave # 3 as FM wave. Most of the candidates defined AM wave as 'a phenomenon in which the amplitude of the carrier is varied while its frequency remains constant' and FM wave as 'a phenomenon in which the frequency of the carrier wave is varied while its amplitude remains constant'.

Example:


a.

Amplitude modulation ⇒ 2.

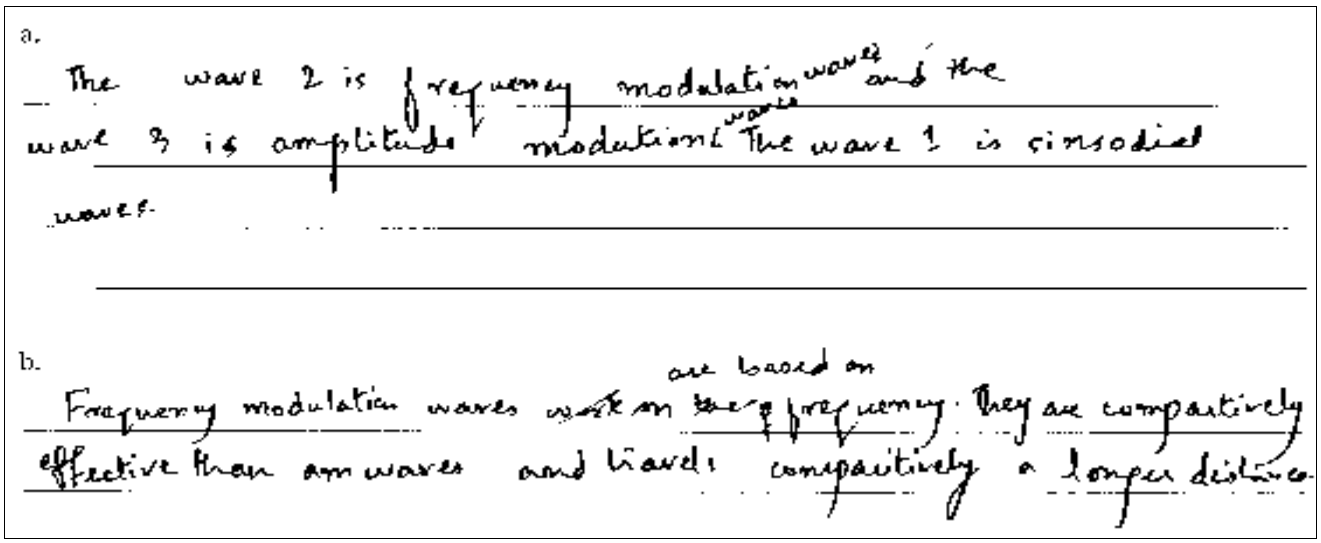
Frequency modulation ⇒ 3.

b.

In AM waves, the amplitude of the carrier waves is modulated according to the amplitude of message, while its frequency remains the same.

Weaker responses revealed that candidates were unable to identify and define the different types of waves. Few of the low scoring candidates identified wave # 2 as FM wave and wave # 3 as AM wave. Similarly, they wrote about different characteristics of waves while defining the different types of waves; like, frequency, wavelength, comparability, density of air, distance travelled by a wave etc.

Example:


a.

The wave 2 is frequency modulation wave and the wave 3 is amplitude modulation wave. The wave 1 is sinsodial wave.

b.

Frequency modulation waves are based on high frequency. They are comparitively effective than am waves and travels comparitively a longer distance.

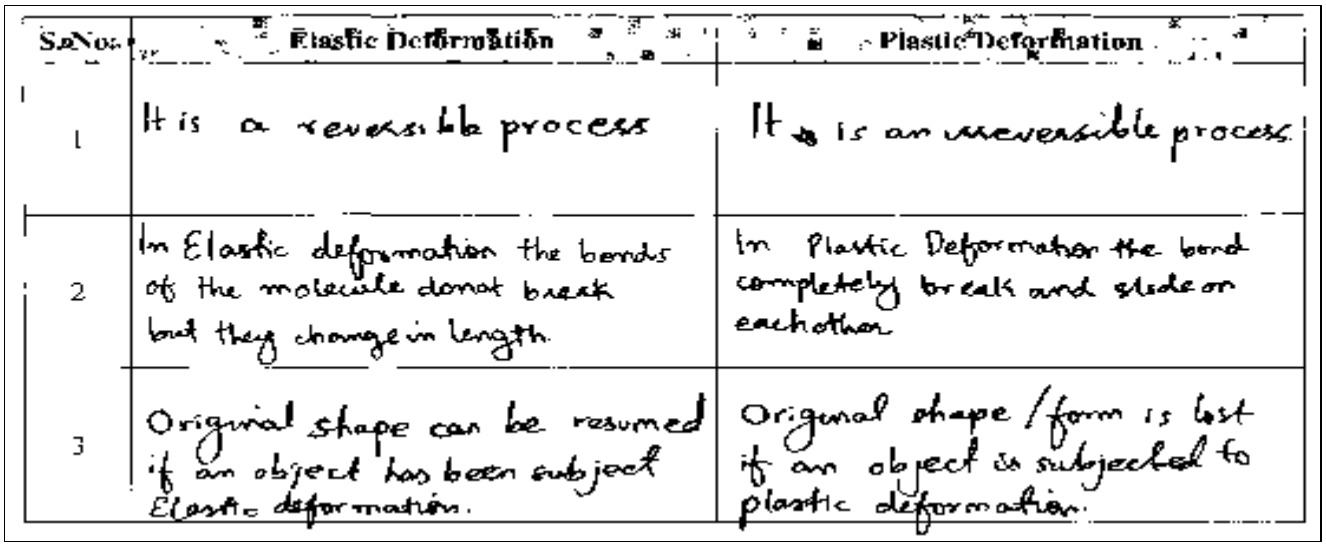
Question 7:

Give THREE differences between elastic and plastic deformations in solids.

Most candidates attempted this question well.

Better responses presented all three correct differences between elastic and plastic deformations in solids by mentioning about their structures, shapes, bonds strength, relationships, either regain their original shape or not and role of heat in their deformation.

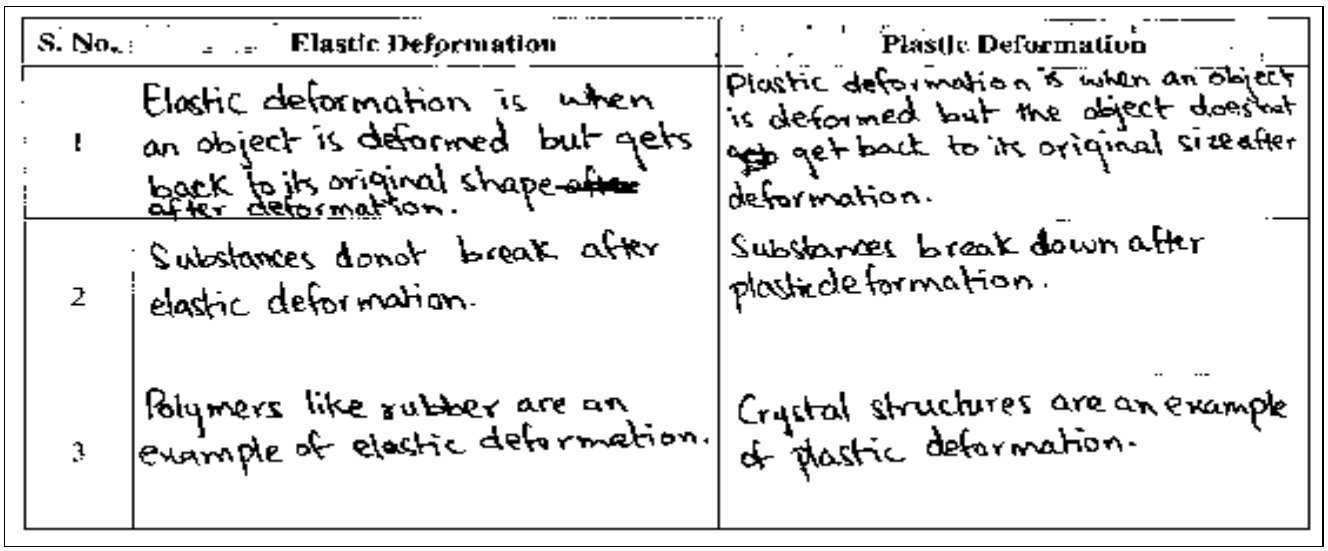
Example:



S.No.	Elastic Deformation	Plastic Deformation
1	It is a reversible process	It is an irreversible process
2	In Elastic deformation the bonds of the molecule donot break but they change in length.	In Plastic Deformation the bond completely break and slide on eachother
3	Original shape can be resumed if an object has been subject Elastic deformation.	Original shape / form is lost if an object is subjected to plastic deformation.

Weaker responses exhibited that candidates were unable to write all three correct differences between elastic and plastic deformations. Few of the low scoring candidates repeated the same idea/ difference in two different points. Therefore, failed to get full marks in their answers.

Example:



S. No.	Elastic Deformation	Plastic Deformation
1	Elastic deformation is when an object is deformed but gets back to its original shape after deformation.	Plastic deformation is when an object is deformed but the object doesnot get back to its original size after deformation.
2	Substances donot break after elastic deformation.	Substances break down after plastic deformation.
3	Polymers like rubber are an example of elastic deformation.	Crystal structures are an example of plastic deformation.

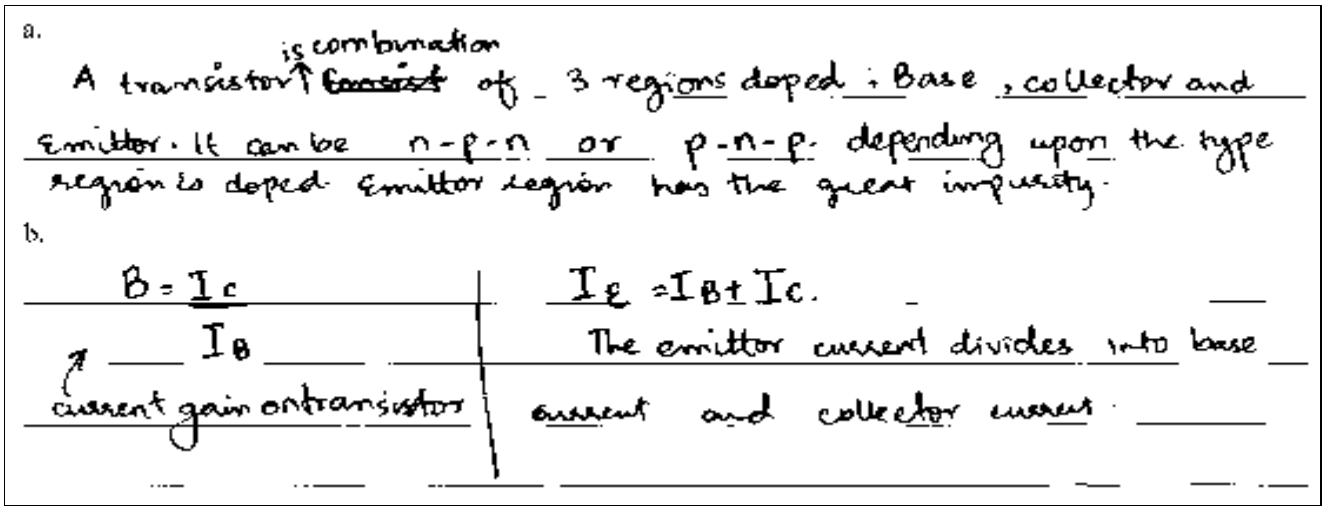
Question 8:

a. Define a transistor.
b. Mention TWO equations which are regarded as fundamental equations of all transistors.

Most candidates performed well in this question.

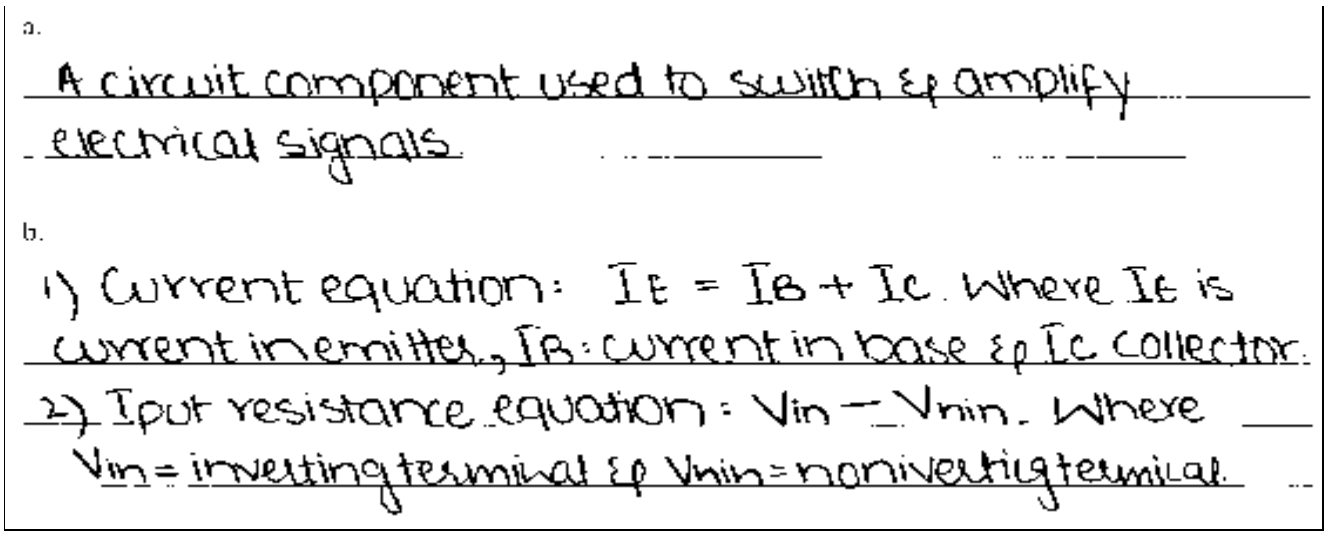
Better responses correctly defined a transistor as a semiconductor device consisting of three electrodes, namely emitter, base and collector. High scoring candidates also correctly wrote two equations which are regarded as fundamental equations of all transistors, i.e, $I_E = I_C + I_B$ and $\beta = \frac{I_C}{I_B}$.

Example:



Weaker responses were successful in writing only one equation which is regarded as fundamental equation of all transistors but were unable to define a transistor correctly. Low scoring candidates wrote about important components of an electric circuit, like, power switch, resistor, capacitor, semiconductor devices etc., which were totally irrelevant to the demand of the question.

Example:



Question 9:

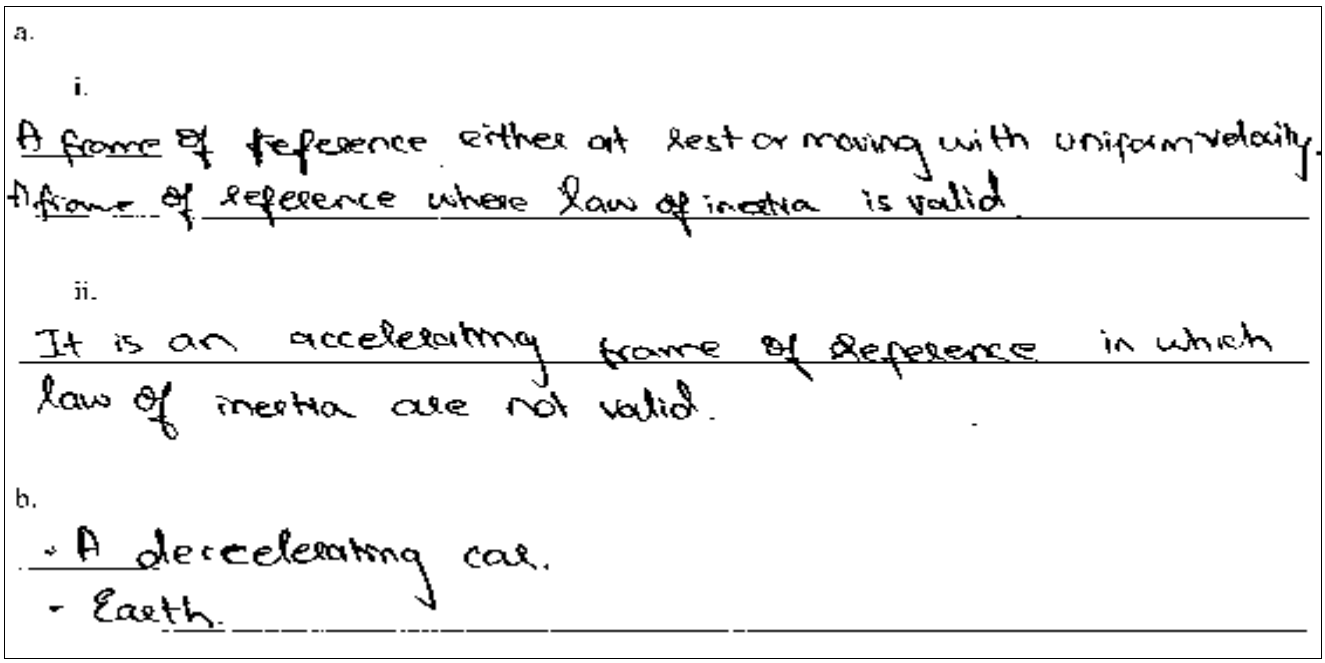
a. Define the following:
   i. Inertial frame of references
   ii. Non-inertial frame of references

b. Give one example of a non-inertial frame of reference.

The question was generally well-attempted.

Better responses correctly defined inertial and non-inertial frame of references and also provided the correct example of non-inertial frame of reference. Below is the example of such kind of better response.

Example:



Weaker responses indicated that candidates were unable to write the correct definitions and were also failed to write one correct example of non-inertial frame of reference. Low scoring candidates wrote about the state of an object i.e., motion and rest in inertial and non-inertial frame of references. They also wrote the same state of an object in the example of a non-inertial frame of reference.

Example:

a.

i.

Frame of reference which is to be compared is when in motion or action. (moving)

ii.

Frame of reference which is to be compared is when static or not moving

b.

A ball at rest is observed by a static person (at rest).

Question 10:

Determine the wavelength of radiation emitted by a hydrogen atom when an electron jumps from n = 4 to the n = 2 orbit in the Bohr atomic model.
(**Note:** Given that Rydberg's constant = R = 1.0974×10^7 m^{-1}.)

Better responses correctly determined the wavelength of radiation emitted by a hydrogen atom. High scoring candidates extracting the correct data from the question, putting the correct formula of wavelength, $\frac{1}{\lambda} = R\left(\frac{1}{p^2} - \frac{1}{n^2}\right)$, substituted the correct values in the formula and finally, calculated the correct answer, $\lambda = 4.859 \times 10^{-7}$ m.

Example:

Data:-

higher orbit = 4 (n)

lower orbit = 2 (p)

$R_H = 1.0974 \times 10^7\ m^{-1}$

$\lambda = ?$

Solution

$\frac{1}{\lambda} = R_H\left[\frac{1}{p^2} - \frac{1}{n^2}\right]$

$\frac{1}{\lambda} = (1.0974 \times 10^7 m)\left[\frac{1}{2^2} - \frac{1}{4^2}\right]$

$\frac{1}{\lambda} = (1.0974 \times 10^7)\left[\frac{1}{4} - \frac{1}{16}\right]$

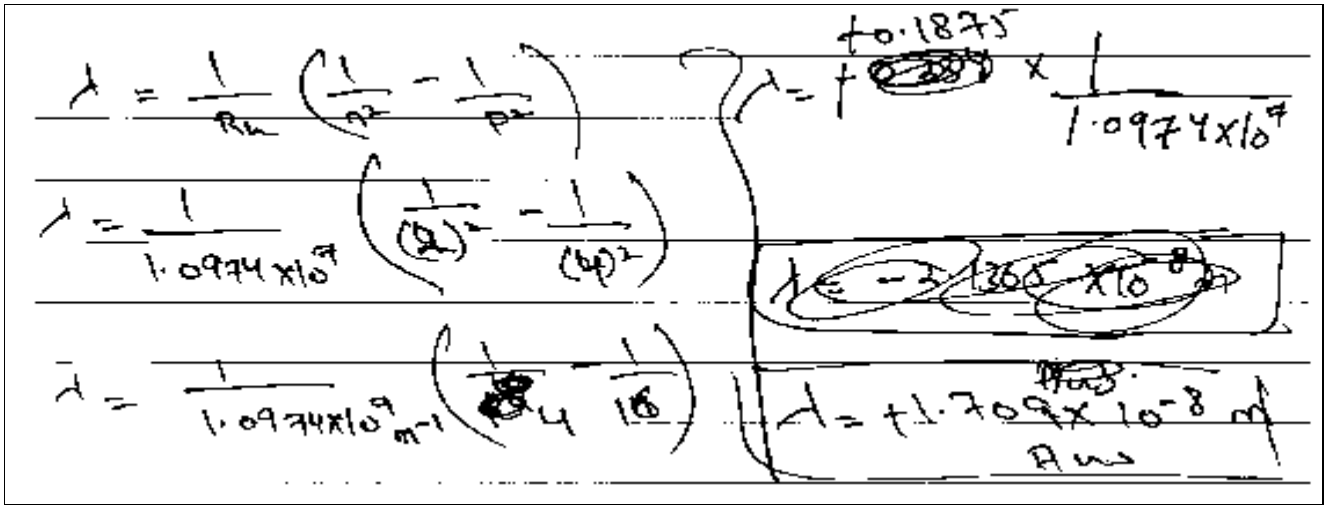
$\frac{1}{\lambda} = (1.0974 \times 10^7)\ [0.1875\ ]$

$\frac{1}{\lambda} = 2057625\ m^{-1}$

$\lambda = 4.859 \times 10^{-7}\ m$

Weaker responses wrote the correct formula of wavelength of radiation emitted by a hydrogen atom but were unable to substitute the correct values in it and got wrong answer. Therefore, lost marks in the correct calculating value of the wavelength.

Example:



Question 11:

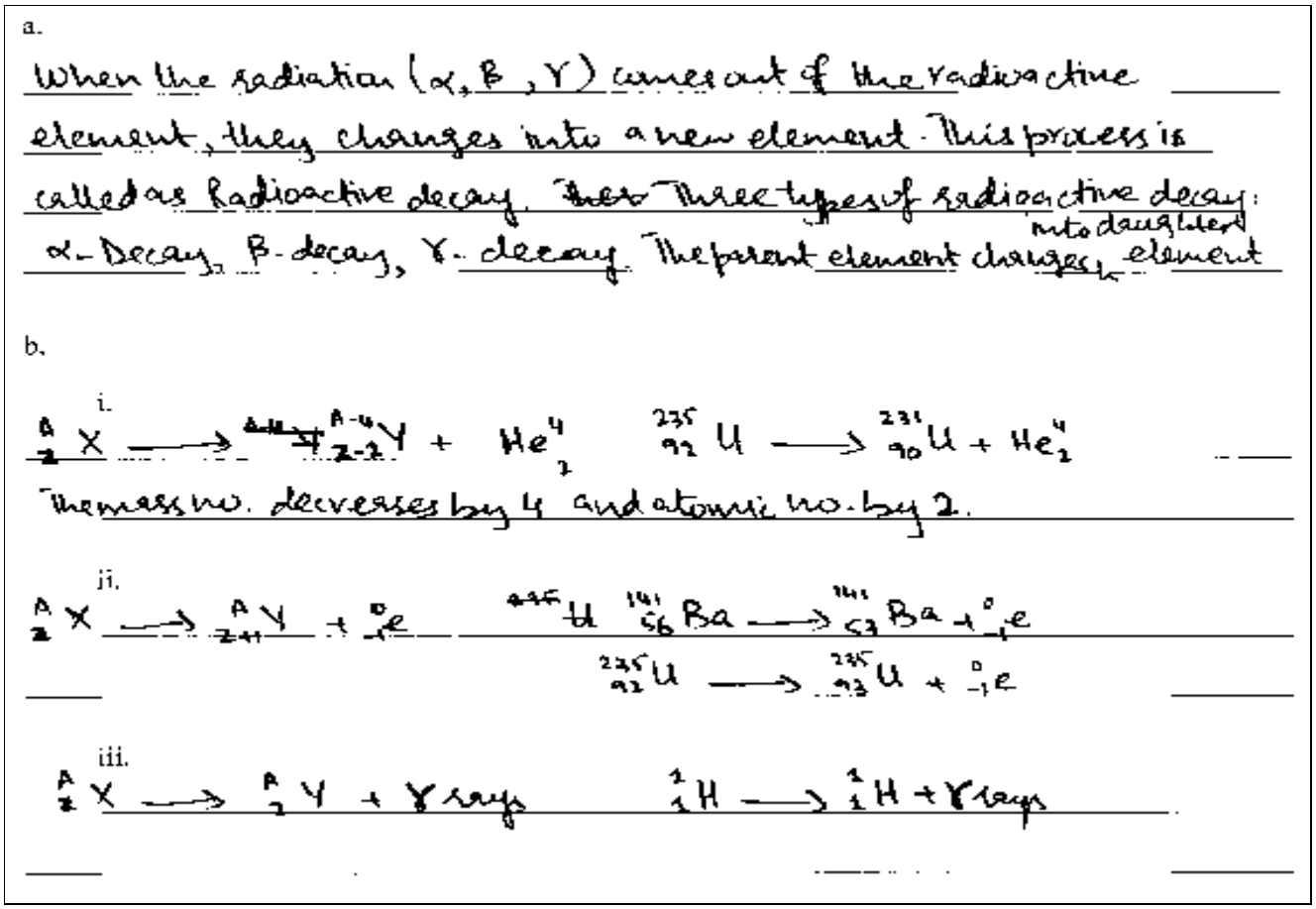
a. Describe the phenomenon of radioactive decay.
b. Write ONE balanced equation for each of the following decays.

   i. Alpha (α)
   ii. Beta (β)
   iii. Gamma (γ)

Most candidates attempted this question well.

Better responses correctly described the phenomenon of radioactive decay along with one balanced equation each for alpha, beta and gamma. One of this kind of answer is shown in the below example of better response.

Example:



Weaker responses incorrectly described the phenomenon of radioactive decay. Few of the low scoring candidates wrote that radioactive decay only occurs in the nuclear fission or fusion reaction. They were also unable to write all the balanced equations for the nuclear decays that showed lack of in-depth understanding of the candidates on this very important topic in nuclear physics.

Example:

a.

It is defined as the radioactivity which is deacaying after the nuclear fission. when value is less than 82 then it is called radioactive decay.

b.

i.

$e = {}^{6}_{4}e + {}^{4}_{2}e$

ii.

$^{213}_{84}U = {}^{213}_{84}U \rightarrow {}^{212}_{82}U$

iii.

$^{1}_{0}l \rightarrow {}^{1}_{1}l + l$

Extended Response Questions (ERQs)
These questions offered a choice between part **a** and **b**

(**Note**: Majority of the candidates attempted Question **12b** and they scored well in this part.)

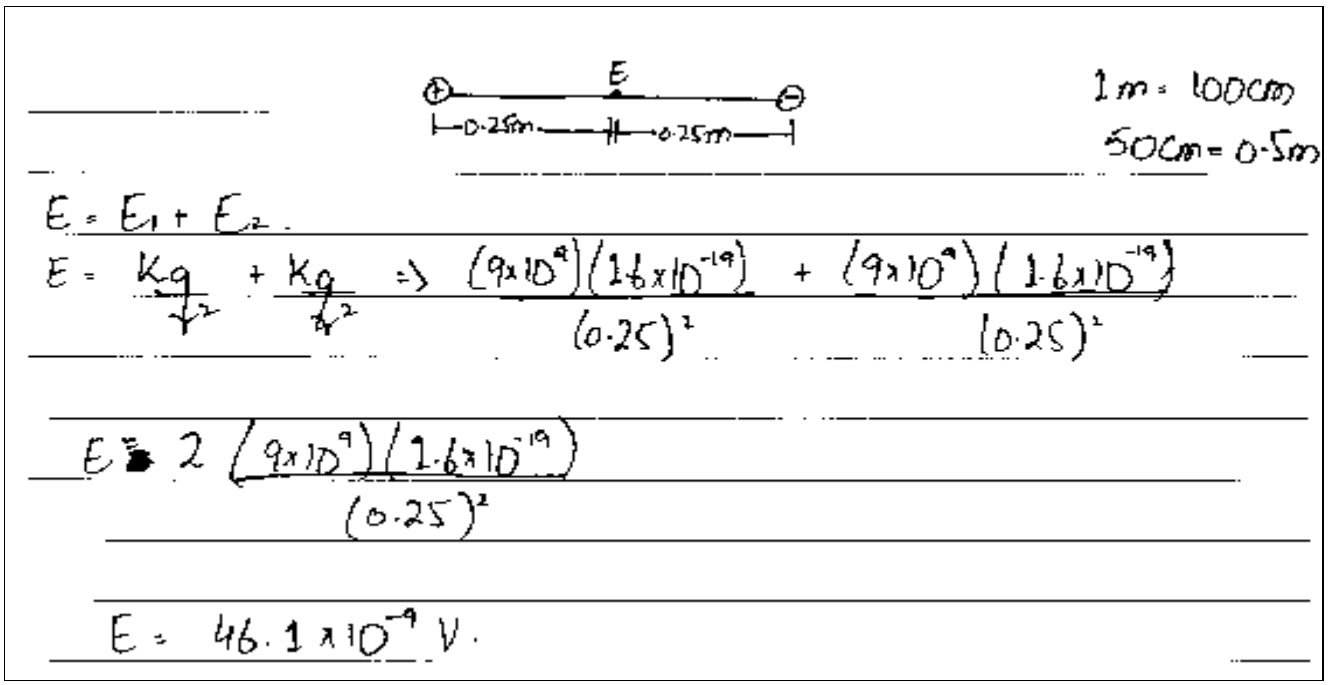
Question 12a:

Calculate the electric field intensity at midpoint between two charges + 1.6×10^{-19} C and -1.6×10^{-19} C, separated by a distance of 50 cm.
(**Note:** The value of the constant of proportionality 'k' = 9×10^9 Nm^2/C^2)

Better responses correctly calculated the electric field intensity by first converting the distance between two charges from 'cm' to 'm', putting the correct formula, $E = \left(\frac{1}{4\pi\pi_o} \times \frac{q_1}{r^2}\right)_1 + \left(\frac{1}{4\pi\pi_o} \times \frac{q_2}{r^2}\right)_2$ and finally, found the correct value of electric field intensity,

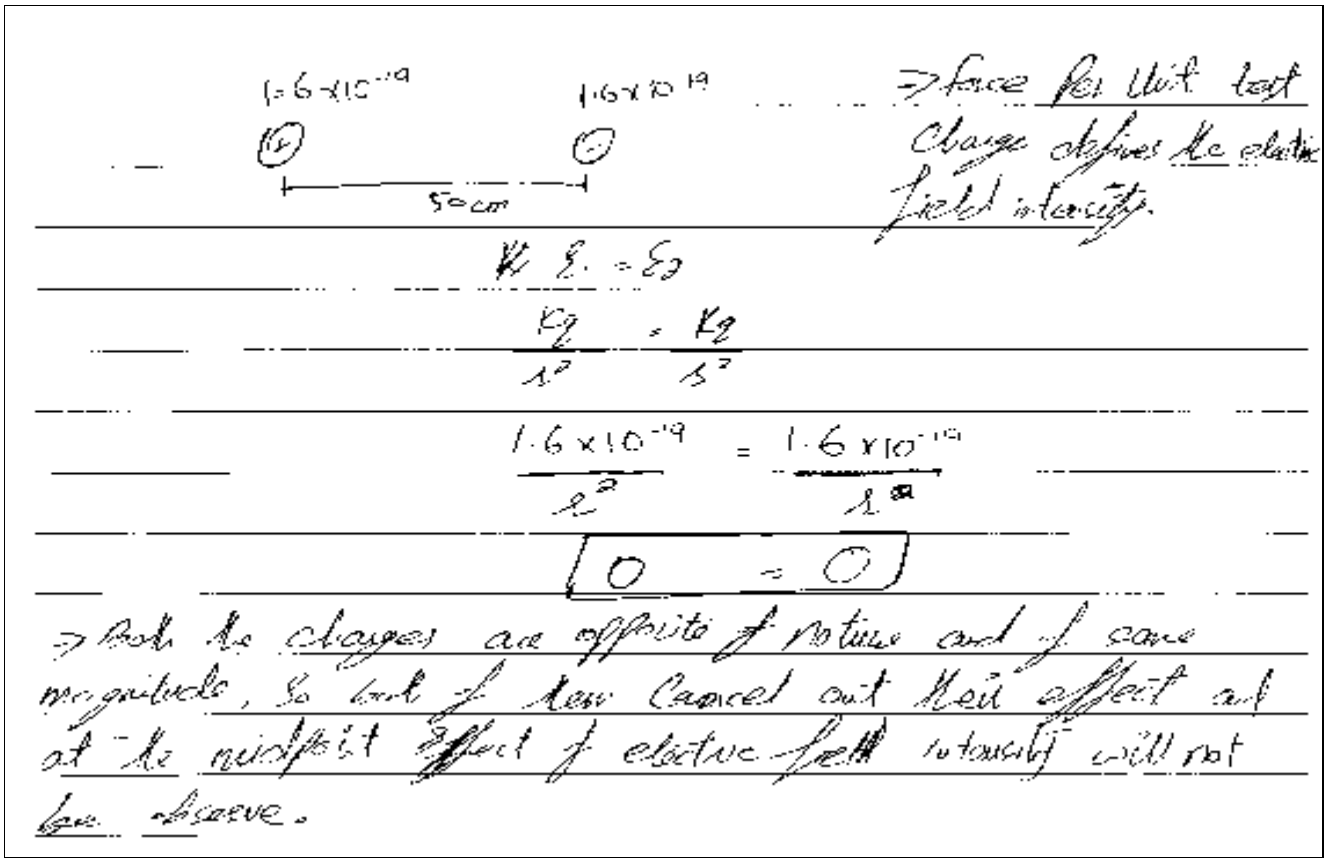
$E = 4.608 \times 10^{-8}$ N/C.

Example:



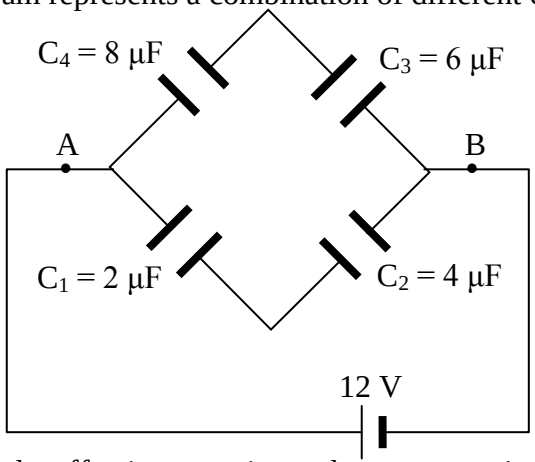
Weaker responses incorrectly calculated the value of electric field intensity. Some of the low scoring candidates made the correct imaginary diagram according to the demand of the question but, they compared both the charges by applying the electric field intensity formulae and got the answer, zero. One of this kind of answer is shown in the below example of weaker response image.

Example:



Question 12b:

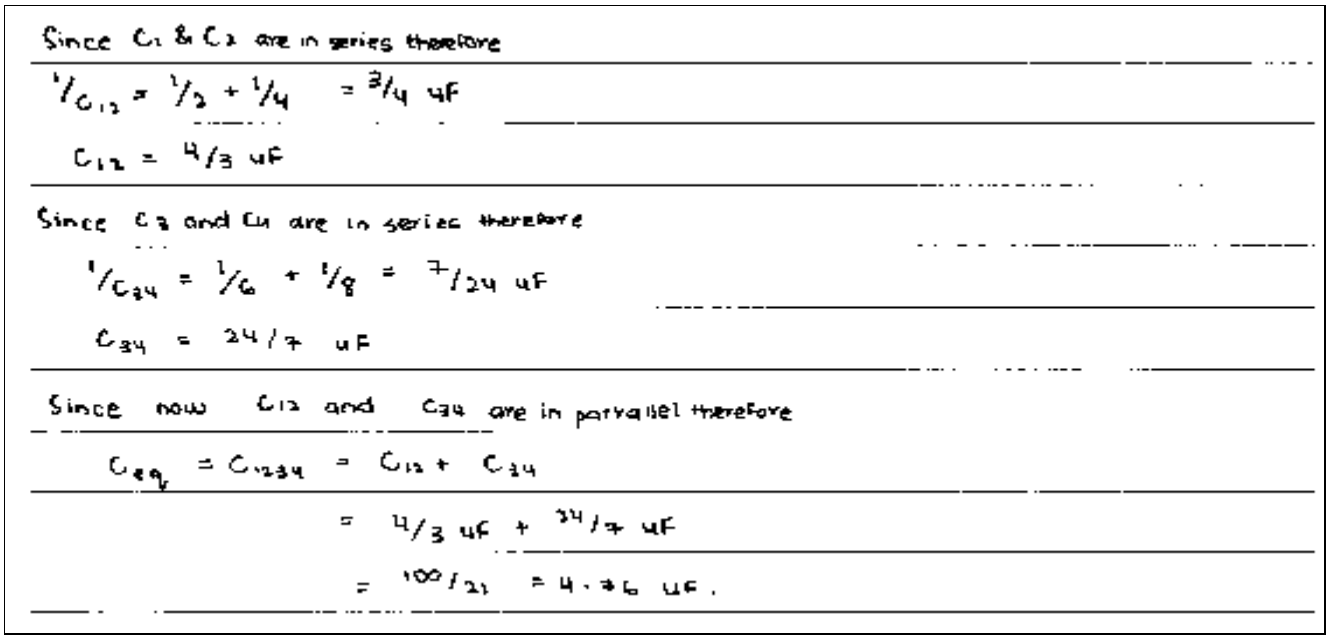
The diagram represents a combination of different capacitors.



Calculate the effective capacitance between terminals (A) and (B).

Better responses correctly calculated the effective capacitance by putting the correct formula in series and parallel combinations simultaneously, i.e. $\frac{1}{C'}=\frac{1}{C_1}+\frac{1}{C_2}$ and $C_e = C'+C''$. High scoring candidates also found the correct values of the effective capacitance in different combinations of capacitors shown in the circuit diagram.

Example:


Since C_1 & C_2 are in series therefore

$1/C_{12} = 1/2 + 1/4 = 3/4$ uF

$C_{12} = 4/3$ uF

Since C_3 and C_4 are in series therefore

$1/C_{34} = 1/6 + 1/8 = 7/24$ uF

$C_{34} = 24/7$ uF

Since now C_{12} and C_{34} are in parallel therefore

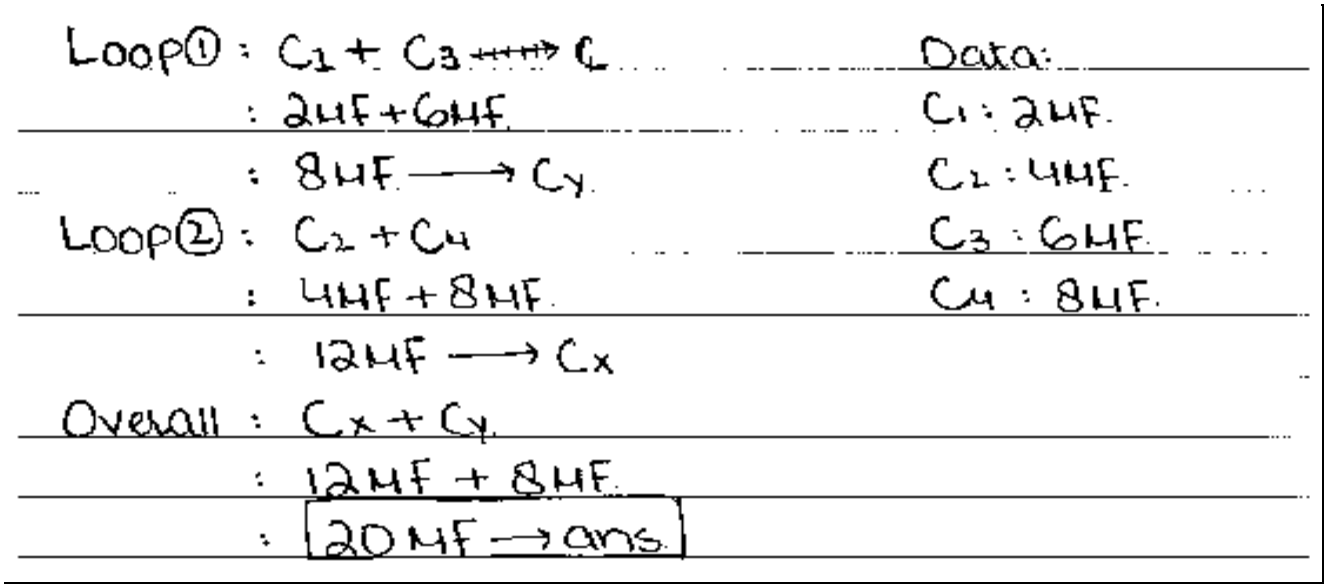
$C_{eq} = C_{1234} = C_{12} + C_{34}$

$= 4/3$ uF $+ 24/7$ uF

$= 100/21 = 4.76$ uF.

Weaker responses reflected that candidates were unable to understand the circuit diagram given in the question. They considered all the capacitors in parallel combination and applied the formula accordingly which shows that candidates did not have any idea about the different combinations of the capacitors or they did not have enough practice in the classrooms. One of this weak response is shown in the below image.

Example:


Loop①: $C_1 + C_3 \Longrightarrow C$

: 2MF + 6MF

: 8MF ⟶ C_y

Loop②: $C_2 + C_4$

: 4MF + 8MF

: 12MF ⟶ C_x

Overall: $C_x + C_y$

: 12MF + 8MF

: 20MF ⟶ ans

Data:

C_1 : 2MF.

C_2 : 4MF.

C_3 : 6MF.

C_4 : 8MF.

(**Note**: Majority of the candidates attempted Question **13b** and they scored well in this part.)

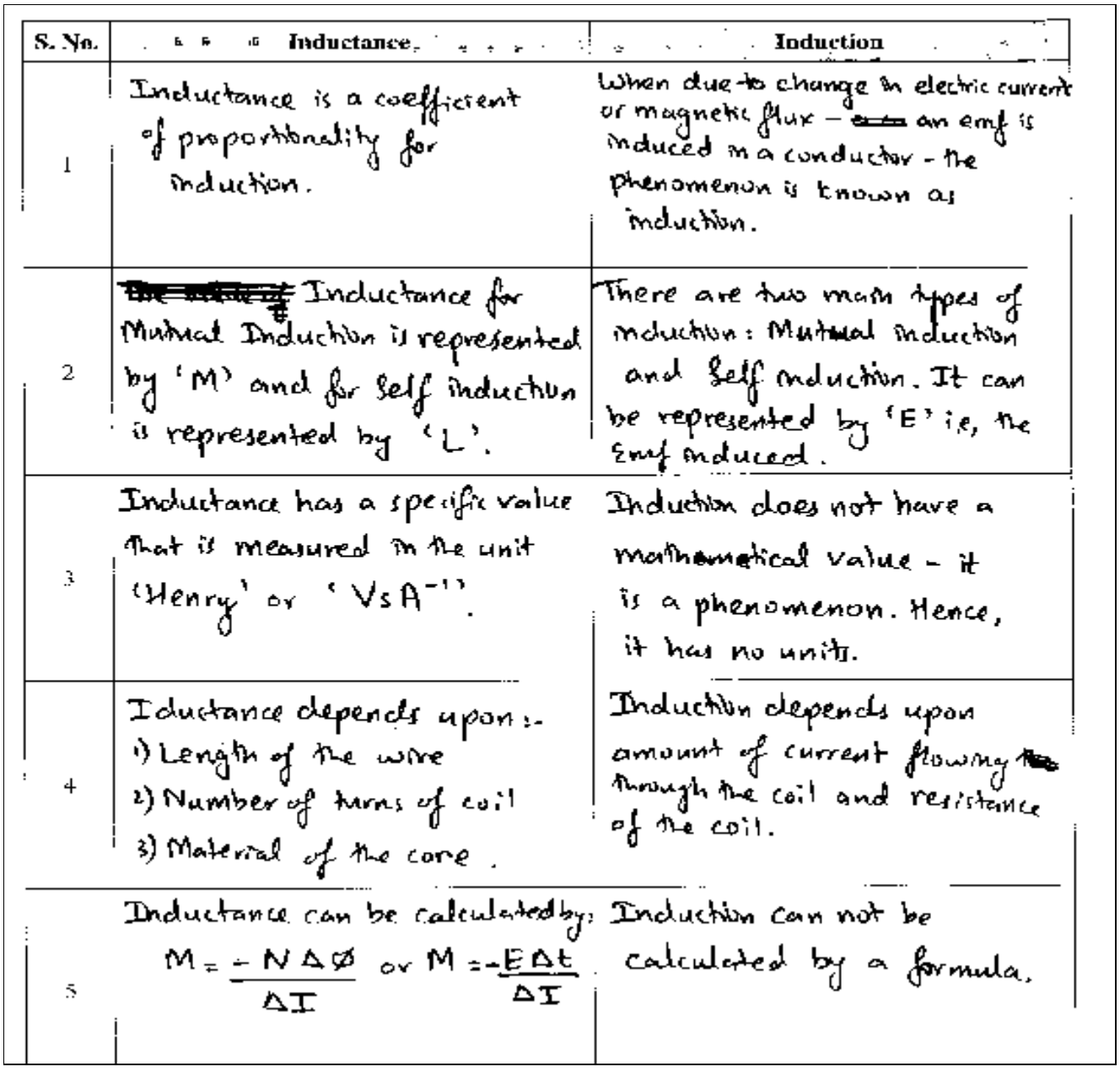
Question 13a:

Write any FIVE differences between inductance and induction.

Most candidates found this question challenging and hence, the performance was below expectations.

Better responses correctly wrote five differences between inductance and induction. Like, the one shown in the below example of better response image.

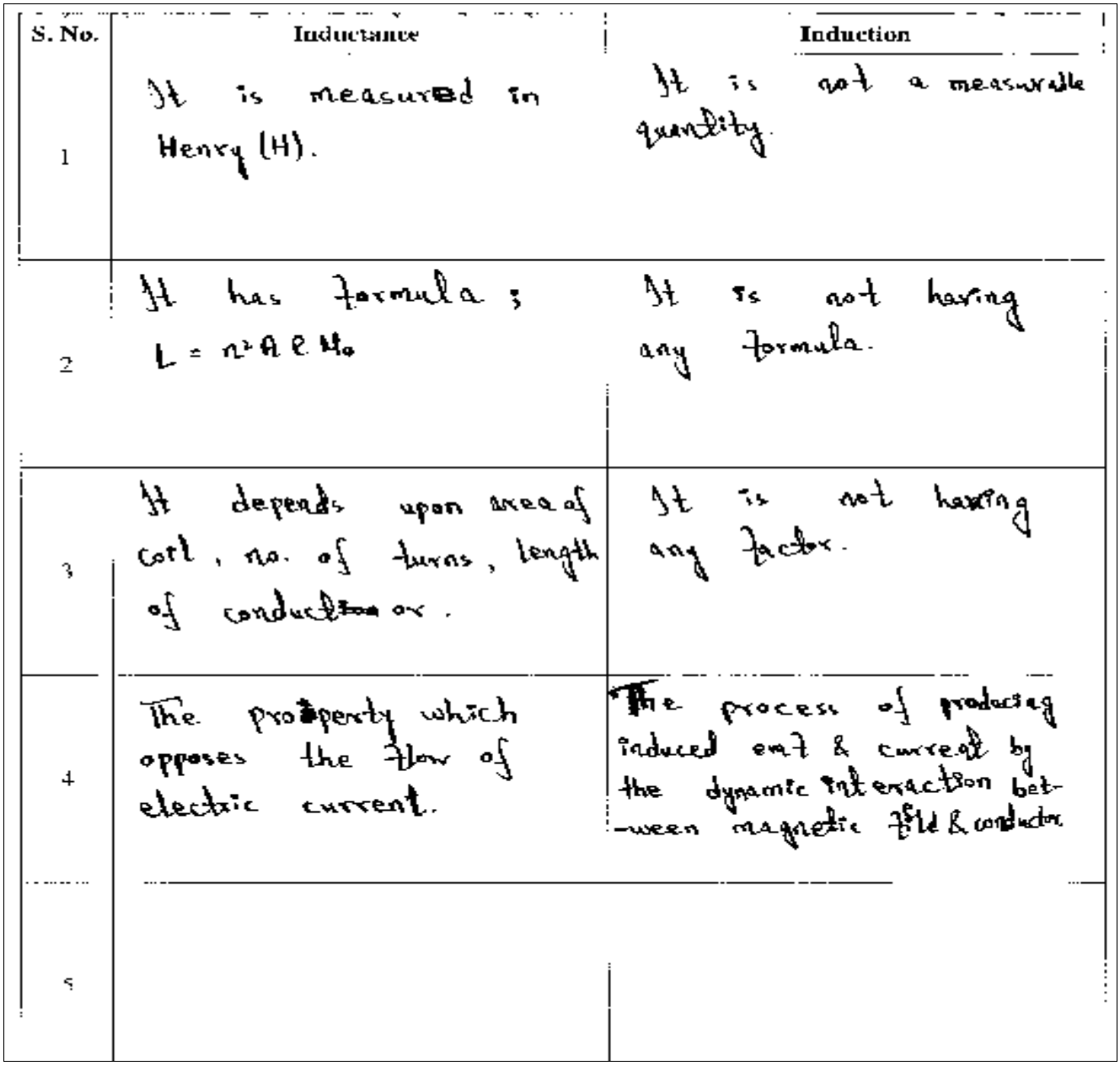
Example:



S. No.	Inductance	Induction
1	Inductance is a coefficient of proportionality for induction.	When due to change in electric current or magnetic flux – an emf is induced in a conductor - the phenomenon is known as induction.
2	Inductance for Mutual Induction is represented by 'M' and for Self Induction is represented by 'L'.	There are two main types of induction: Mutual induction and Self induction. It can be represented by 'E' i.e, the Emf induced.
3	Inductance has a specific value that is measured in the unit 'Henry' or 'VsA^{-1}'.	Induction does not have a mathematical value - it is a phenomenon. Hence, it has no units.
4	Inductance depends upon :- 1) Length of the wire 2) Number of turns of coil 3) Material of the core.	Induction depends upon amount of current flowing through the coil and resistance of the coil.
5	Inductance can be calculated by: $M = \frac{-N\Delta\phi}{\Delta I}$ or $M = \frac{-E\Delta t}{\Delta I}$	Induction can not be calculated by a formula.

Weaker responses showed candidates' weak knowledge about inductance and induction. Most of the low scoring candidates were unable to write all five differences and repeated the same points at different places; like, number of turns and length of the coil, unit of inductance and induction, flow of charges etc. Some of the candidates were confused between inductance and induction.

Example:



S. No.	Inductance	Induction
1	It is measured in Henry (H).	It is not a measurable quantity.
2	It has formula ; $L = n^2 A \ell \mu_0$	It is not having any formula.
3	It depends upon area of cort, no. of turns, length of conductor.	It is not having any factor.
4	The property which opposes the flow of electric current.	The process of producing induced emf & current by the dynamic interaction between magnetic field & conductor
5		

Question 13b:

Write any FIVE differences between alternating and direct current generators.

Most candidates attempted this question well.

Better responses correctly wrote all five differences between alternating and direct current generators. One of the example of better response is shown in the below image.

Example:

S. No.	AC generator	DC generator
1	→ AC generators produce output current which is alternating in nature i.e it keeps on changing its direction	→ DC generators produce output current which is direct current i.e it is unidirectional.
2	→ AC generators require higher voltages i.e around 120 V for their operation.	→ DC generators require less voltage about 50 V for their operation
3	→ AC generators have slip rings ~~also~~ attached to the two terminals of the coil	→ DC generators have split rings which acts as commutator preventing the direction of current from being changed.
4	→ AC generators donot have steady magnetism along the wire i.e the magnet also rotates with the rotating coil.	→ DC generator have steady magnetism along the wire.
5	→ AC generators are usually used at homes in operating smaller electric devices / appliances.	→ DC generators are usually used to operate heavy motors and machinery in industries etc.

Weaker responses failed to write all the correct five differences. Some of the low scoring candidates wrote totally irrelevant points; like, positive and negative poles, lower and higher potential, induced current and electromotive current, abbreviations of alternating current and direct current etc.

Example:

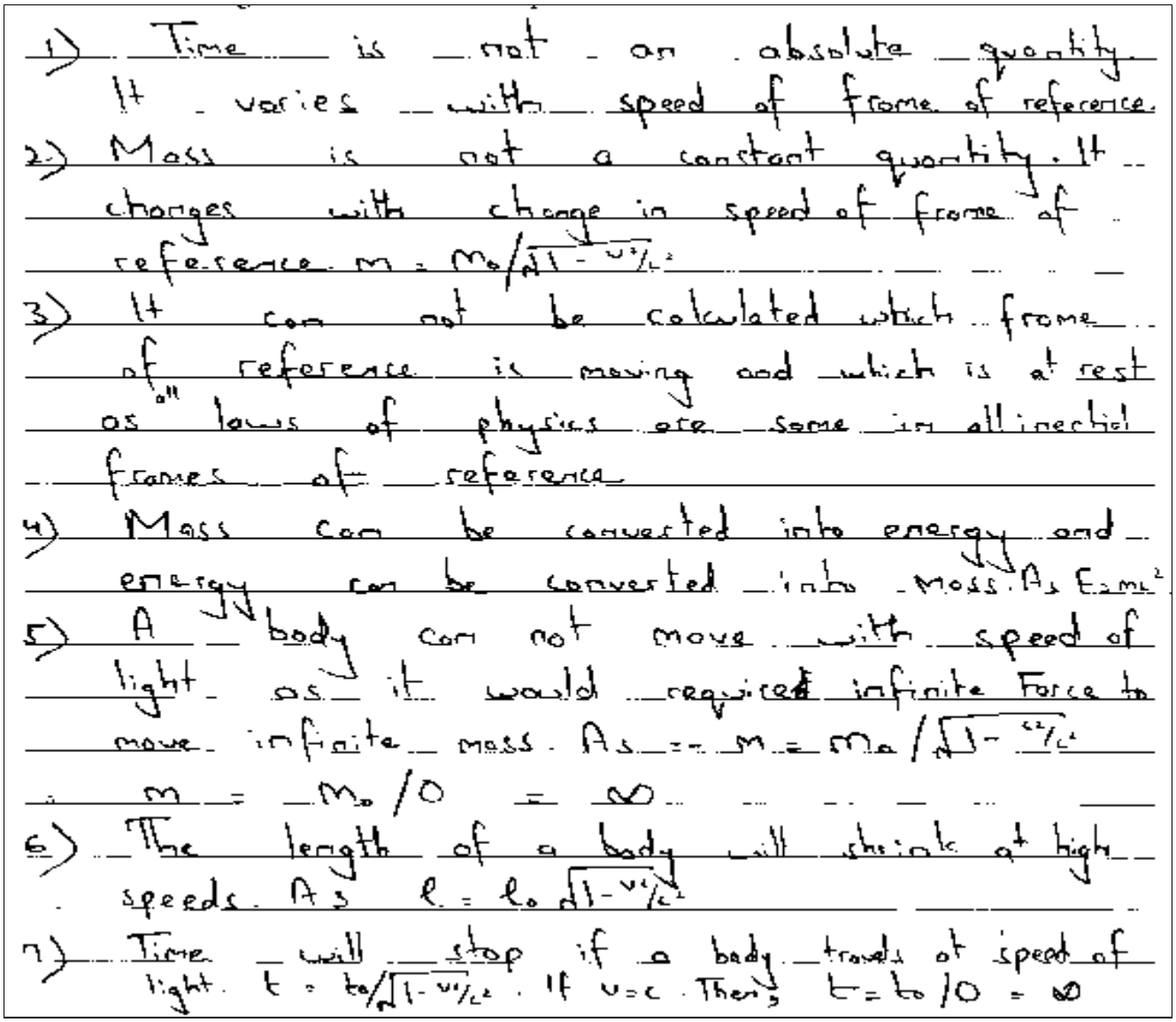
S. No.	AC generator	DC generator
1	it seems positive to negative.	it seems negative to positive.
2	higher potenial to lower potenial	lower potenial to higher potenial
3	it convert e.m.f current into induced current	it convert induced current into e.m.f current.
4	Ac generator is commonly use in circuits	Dc generator can'not we commonly in circuits.
5	Ac is Alternating current	Dc is direct current.

(**Note**: Majority of the candidates attempted Question **14a** and they scored well in this part.)

Question 14a:
Describe any FIVE conclusions that can be extracted by Einstein's special theory of relativity.

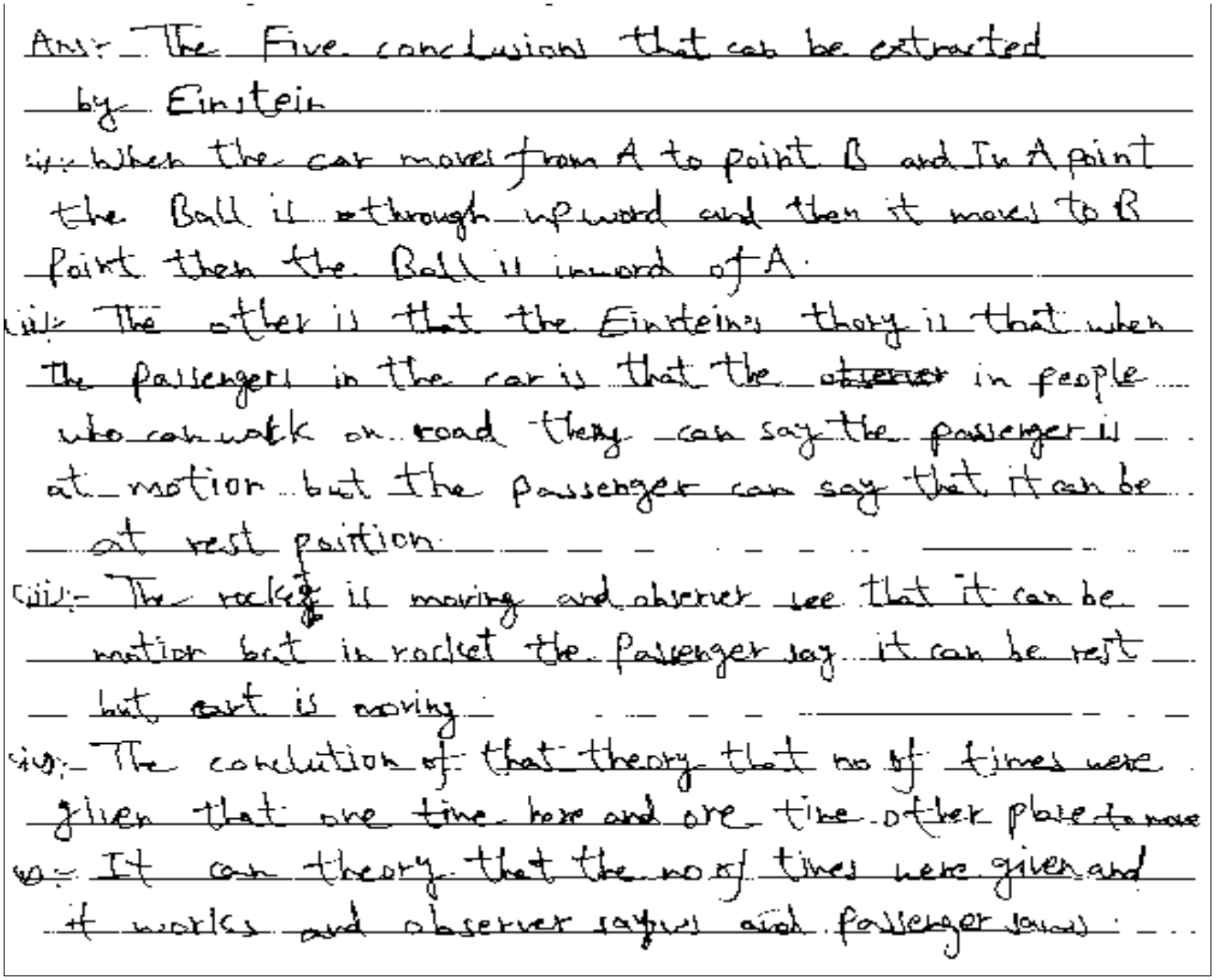
Better responses correctly described all the conclusions that can be extracted by Einstein's special theory of relativity. High scoring candidates described their answers in terms of relative motions of the different objects, time and space relationship, four dimensions, mass increment, length contraction, time dilation, mass-energy equation and with respect to the speed of light.

Example:


1) Time is not an absolute quantity. It varies with speed of frame of reference.
2) Mass is not a constant quantity. It changes with change in speed of frame of reference. $m = m_0/\sqrt{1 - v^2/c^2}$
3) It can not be calculated which frame of reference is moving and which is at rest as all laws of physics are same in all inertial frames of reference
4) Mass can be converted into energy and energy can be converted into mass. As $E = mc^2$
5) A body can not move with speed of light as it would required infinite Force to move infinite mass. As $m = m_0/\sqrt{1 - v^2/c^2}$ $m = m_0/0 = \infty$
6) The length of a body will shrink at high speeds. As $l = l_0\sqrt{1 - v^2/c^2}$
7) Time will stop if a body travels at speed of light. $t = t_0/\sqrt{1 - v^2/c^2}$. If $v = c$. Then $t = t_0/0 = \infty$

Weaker responses failed to describe all the conclusions that can be extracted by Einstein's special theory of relativity. Few of the low scoring candidates wrote about the different states of an object (ball, rocket, cart, passenger etc.) in a spacecraft. Furthermore, candidates also wrote about black holes, time machines and space and its limitation which are irrelevant to the demand of the question.

Example:


Ans:- The Five conclusion that can be extracted by Einstein
(i):- When the car moves from A to point B and In A point the Ball is thrown upward and then it moves to B point then the Ball is inword of A.
(ii):- The other is that the Einstein's theory is that when the passengers in the car is that the observer in people who can walk on road they can say the passenger is at motion but the passenger can say that it can be at rest position
(iii):- The rocket is moving and observer see that it can be motion but in rocket the passenger say it can be rest but cart is moving
(iv):- The conclution of that theory that no of times were given that one time here and one time other place to note
(v):- It can theory that the no of times were given and it works and observer says this passenger says

Question 14b:

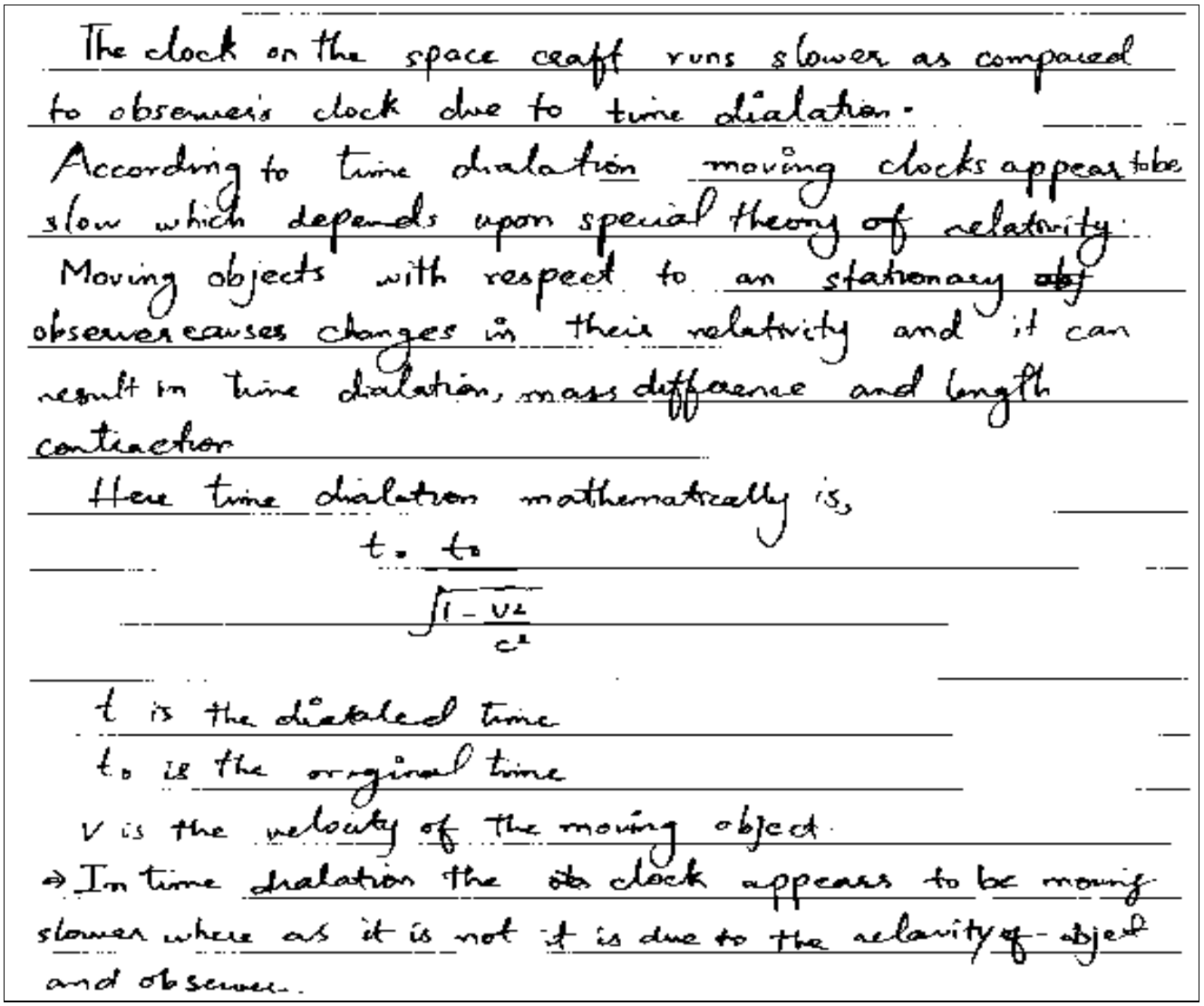
Suppose a spacecraft is moving from the Earth to a distant star. A stationary observer is observing the spacecraft from the Earth. If the spacecraft and the observer have similar clocks, then according to the special theory of relativity, the clock on the spacecraft runs slower as compared to the observer's clock.

Elaborate the given scenario with the help of mathematical formulas.

Most candidates were not able to perform well in this question.

Better responses correctly elaborated the given scenario with the help of detailed description and mathematical expressions. High scoring candidates wrote about the description and mathematical expressions in terms of time dilation, a clock at spacecraft will count less number of seconds than a clock on earth and the time interval between two events also they wrote $t = \frac{t_o}{\sqrt{1-\frac{v^2}{c^2}}}$ and $\sqrt{1-\frac{v^2}{c^2}}$.

Example:



The clock on the space craft runs slower as compared to observer's clock due to time dialation.
According to time dialation moving clocks appear to be slow which depends upon special theory of relativity.
Moving objects with respect to an stationary ~~obj~~ observer causes changes in their relativity and it can result in time dialation, mass diffaence and length contraction
Here time dialation mathematically is,

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

t is the dialated time
t_0 is the original time
v is the velocity of the moving object.
⇒ In time dialation the ~~its~~ clock appears to be moving slower where as it is not it is due to the relativity of object and observer.

Weaker responses correctly wrote either both mathematical expressions or one. Most of the low scoring candidates wrote about the concept of relative motion, its consequences, frames of reference, time dilation. Few of the candidates mixed up results of special theory of relativity, which are totally irrelevant to the demand of the question and shows candidates lack of understanding and in-depth knowledge on this most important topic of modern physics.

Example:

According to special theory of relativity if a body is moving with very high speed approaching speed of light then the time observed in that body's will dialate which means time will get slower as compared to time observed by an observer who is out out that body. If time in the spacecraft is t and time observed by the observer outside the body is t_0 then relation between time between body and observer can be given by :

$$t = t_0 \sqrt{1 - \frac{c^2}{v^2}}$$